

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?

- **Native** – Built for one operating system (Android or iOS). Best for apps that need high performance and full device features.
- **Hybrid** – Uses web technologies inside a mobile app. Ideal for simple apps with a limited budget.
- **Cross-platform** – Uses one codebase for multiple platforms. Great for faster development while maintaining good performance.

2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?

The **Flutter Ecosystem** is a collection of tools, libraries, widgets, and services that work together to build apps. It helps developers create attractive, fast, and responsive applications for Android, iOS, web, and desktop using a single codebase.

3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?

The **Software Development Life Cycle (SDLC)** is a step-by-step process for planning, creating, testing, deploying, and maintaining software. It helps ensure the application is organized, reliable, and meets user needs.

4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?

Flutter provides the tools to build the app, while the SDLC provides the process to develop it efficiently. Together, they help teams create high-quality applications with fewer errors and smoother project management.

5. Why should developers understand these concepts before starting an application project?

Understanding these concepts helps developers choose the right technology, plan the project effectively, reduce development risks, and build applications that are efficient, scalable, and user-friendly.